Q1. Solve the following questions
Write a script in R to create a list of students and perform the following

i) **Give names to the students in the list**
ii) **Add a student at the end of the list.**
iii) **Remove the first Student.**
iv) **Update the second last student.**

Ans->

```
# Create a list of students
students <- list("Aastha", "Rahul", "Priya", "Kiran")

# i) Give names to the students in the list
names(students) <- c("Student1", "Student2", "Student3", "Student4")
print("Named List:")
print(students)

# ii) Add a student at the end of the list
students[[length(students) + 1]] <- "Neha"
print("After Adding Student:")
print(students)

# iii) Remove the first student
students <- students[-1]
print("After Removing First Student:")
print(students)

# iv) Update the second last student
students[[length(students) - 1]] <- "Updated_Name"
print("After Updating Second Last Student:")
print(students)
```

Q2. Solve the following questions.

A. **Write a R program to find the mean of numbers from 20 to 60**
B. **Write a R program to find the variance of numbers from 101 to 121.**

Ans->

Mean of numbers from 20 to 60

```
numbers <- 20:60
mean_value <- mean(numbers)

print("Mean is:")
print(mean_value)
```

variance of numbers from 101 to 121

```
numbers <- 101:121
variance_value <- var(numbers)
```

print("Variance is:")
print(variance_value)

Q2) Q1. Solve the following questions.
Create the following matrices in R and perform the following operations:

$$\boldsymbol{A} = \begin{matrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{matrix}$$

$$\boldsymbol{B} = \begin{matrix} 15 & 21 & 13 \\ 21 & 10 & 7 \\ 39 & 7 & 30 \end{matrix}$$

A. Addition (A + B)
B. Subtraction (A – B)
C. Multiplication (AB)
D. Scalar Multiplication
i. 21A
ii. 15B
Ans->

```
# Create Matrix A
A <- matrix(c(1,2,3,4,5,6,7,8,9), nrow=3, byrow=FALSE)

# Create Matrix B
B <- matrix(c(15,21,39,21,10,7,13,7,30), nrow=3, byrow=FALSE)

print("Matrix A:")
print(A)

print("Matrix B:")
print(B)

# A. Addition (A + B)
add <- A + B
print("Addition (A + B):")
print(add)

# B. Subtraction (A - B)
sub <- A - B
print("Subtraction (A - B):")
print(sub)

# C. Multiplication (A %*% B)
mul <- A %*% B
print("Multiplication (A * B):")
```

```
print(mul)
```

 # D. Scalar Multiplication
 # i. 21A

```
scalar_A <- 21 * A
print("21 * A:")
print(scalar_A)
```

 # ii. 15B

```
scalar_B <- 15 * B
print("15 * B:")
print(scalar_B)
```

Q2. Solve the following questions.
A. Write a R program to find the mean of numbers from 20 to 60
B. Write a R program to find the sum of numbers from 101 to 121.

Ans->

mean of numbers from 20 to 60

```
numbers <- 20:60
mean_value <- mean(numbers)

print("Mean is:")
print(mean_value)
```

sum of numbers from 101 to 121.

```
numbers <- 101:121
sum_value <- sum(numbers)

print("Sum is:")
print(sum_value
```

Q3. Q1. Solve the following questions.
A. Write an R program to convert a given matrix to a list and print list in ascending order.

Matrix = 1 54 1 3
21 10 7
39 23 41

Ans->
 # Create Matrix F

```
F <- matrix(c(1,21,39,54,10,23,13,7,41), nrow=3, byrow=FALSE)
print("Matrix F:")
print(F)
# Convert matrix to list
list_F <- as.list(F)
```

```
print("Matrix converted to list:")
print(list_F)
# Convert to vector and sort in ascending order
sorted_values <- sort(unlist(list_F))
print("List in ascending order:")
print(sorted_values)
```

B. Write a R program to multiply the matrix by a scalar. Find the value of 21F.

```
# Multiply matrix by scalar 21
result <- 21 * F
print("21 * F:")
print(result)
```

Q2. Solve the following questions.
A. Print the version of R installation.
B. Write a R program to merge two given lists into one list.
List 1 : (5, hello, 23, data, [1, 2, 3, 4, 5], science, 1:5)
List 2: (a1, a2, a3, 9, 7, 3)

Ans->
A)

```
# Print R version
print(R.version.string
```

B)

```
# Create List 1
list1 <- list(5, "hello", 23, "data", c(1,2,3,4,5), "science", 1:5)
# Create List 2
list2 <- list("a1", "a2", "a3", 9, 7, 3)
# Merge lists
merged_list <- c(list1, list2)
print("Merged List:")
print(merged_list)
```

Q4) Q1. Solve the following questions.

A. Write R program to read number taken from the user and print corresponding day name in a week.
B. Write R program to find whether given number is positive or negative. Given Numbers : 21, - 27, 13

Ans->

```
# Take input from user

num <- as.integer(readline(prompt = "Enter a number (1-7): "))
```

```
# Check and print day
if(num == 1) {
  print("Sunday")
} else if(num == 2) {
  print("Monday")
} else if(num == 3) {
  print("Tuesday")
} else if(num == 4) {
  print("Wednesday")
} else if(num == 5) {
  print("Thursday")
} else if(num == 6) {
  print("Friday")
} else if(num == 7) {
  print("Saturday")
} else {
  print("Invalid input")
}# Given numbers
numbers <- c(21, -27, 13)
# Loop through numbers
for (num in numbers) {
  if (num > 0) {
    print(paste(num, "is Positive"))
  } else if (num < 0) {
    print(paste(num, "is Negative"))
  } else {
    print(paste(num, "is Zero"))
  }
```

}

Q2. Solve the following questions.
Write a R program to find the covariance of numbers from 20 to 60

Ans->
Find covariance of numbers from 20 to 60

```
# Create sequence from 20 to 60
x <- 20:60
# Create another variable (example: y = x^2)
y <- x^2
# Calculate covariance
covariance <- cov(x, y)
# Print result
print(paste("Covariance is:", covariance))
```

Q5. Solve the following questions.
Create the following matrices in R and perform the following operations:

$$\boldsymbol{E} = \begin{matrix} 11 & 44 & 77 \\ 22 & 55 & 88 \\ 33 & 66 & 99 \end{matrix}$$

$$\boldsymbol{F} = \begin{matrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{matrix}$$

B. Subtraction (E – F)
C. Multiplication (EF)
D. Scalar Multiplication
i. 51E ii. 13F
Ans->

```
# Create Matrix E
E <- matrix(c(11, 22, 33,
              44, 55, 66,
              77, 88, 99),
            nrow = 3, byrow = FALSE)

# Create Matrix F
F <- matrix(c(1, 2, 3,
              4, 5, 6,
              7, 8, 9),
            nrow = 3, byrow = FALSE)

# Display matrices
print("Matrix E:")
print(E)

print("Matrix F:")
```

print(F)

 # A. Addition
add_result <- E + F
print("E + F:")
print(add_result)

 # B. Subtraction
sub_result <- E - F
print("E - F:")
print(sub_result)

 # C. Multiplication (Matrix multiplication)
mul_result <- E %*% F
print("E %*% F:")
print(mul_result)

 # D. Scalar Multiplication
 # i. 51E
scalar_E <- 51 * E
print("51 * E:")
print(scalar_E)

 # ii. 13F
scalar_F <- 13 * F
print("13 * F:")
print(scalar_F)

Q2. Solve the following questions.
A. Write a R program to find the mean of numbers from -20 to +20
B. Write a R program to find the third quartile in the sequence of numbers from 101 to 121.
Ans-> mean of numbers from -20 to +20

```
# Create sequence
nums <- -20:20
# Calculate mean
mean_value <- mean(nums)
print(paste("Mean is:", mean_value))
```

Third quartile in the sequence of numbers from 101 to 121.

```
# Create sequencenums <- 101:121

# Find third quartile

q3 <- quantile(nums, 0.75)

print(paste("Third Quartile (Q3) is:", q3))
```

Q6. Q1. Solve the following questions.
Write a R program to get the unique elements of a given string and unique numbers of vector.
STRING : datascience
VECTOR : 2, 3, 2, 5, 4, 5
Ans->

```
# Given string
str <- "datascience"
# Split string into characters
chars <- unlist(strsplit(str, ""))
# Get unique characters
unique_chars <- unique(chars)
print("Unique characters in string:")
print(unique_chars)
# Given vector
vec <- c(2, 3, 2, 5, 4, 5)
# Get unique numbers
unique_nums <- unique(vec)
print("Unique numbers in vector:")
print(unique_nums)
```

Q2. Solve the following questions. Write a R program to take input from the user and display the values.

A. Name
B. Age
C. Gender
D. Country
Ans->

```
# Taking input from user
name <- readline(prompt = "Enter your Name: ")
age <- readline(prompt = "Enter your Age: ")
gender <- readline(prompt = "Enter your Gender: ")
country <- readline(prompt = "Enter your Country: ")
# Displaying values
print(paste("Name:", name))
print(paste("Age:", age))
print(paste("Gender:", gender))
print(paste("Country:", country))
```

Q7. Q1. Solve the following questions.

Write a script in R to create a list of students and perform the following

i) **Give names to the students in the list.**
ii) **Add a student at the end of the list.**
iii) **Remove the second Student.**
iv) **Update the last student.**

Ans->

```
# Create a list of students
students <- list("John", "Alice", "Bob")
# i) Give names to the students in the list
names(students) <- c("Student1", "Student2", "Student3")
print("Named list:")
print(students)
# ii) Add a student at the end
students[[length(students) + 1]] <- "David"
print("After adding a student:")
print(students)
# iii) Remove the second student
students <- students[-2]
print("After removing second student:")
print(students)
# iv) Update the last student
students[[length(students)]] <- "Emma"
print("After updating last student:")
print(students)
```

Q2. Solve the following questions. Write an R Program to calculate Decimal into binary of a given number. Given Decimal Numbers :

A. 21
B. 13
C. 27
D. 72

Ans->

```
# Function to convert decimal to binary
decimal_to_binary <- function(n) {
  paste(rev(as.integer(intToBits(n))), collapse = "")
}
# Given numbers
numbers <- c(21, 13, 27, 72)
# Convert and print
for (num in numbers) {
  binary <- decimal_to_binary(num)
```

```
  # Remove leading zeros
  binary <- sub("^0+", "", binary)
  print(paste("Decimal:", num, "Binary:", binary))
}
```

Q8. Q1. Solve the following questions.
Write a R program to create a list containing a vector, a matrix and a list and give names to the elements in the list. Access the 1st and 2nd element of the list
Vector : [1, 2, 3, 4, 5]
Matrix :

$$\begin{matrix} \mathbf{1} & \mathbf{4} & \mathbf{7} \\ \mathbf{2} & \mathbf{5} & \mathbf{8} \\ \mathbf{3} & \mathbf{6} & \mathbf{9} \end{matrix}$$

List : (a1, a2, a3)
Ans->

```
# Create vector
vec <- c(1, 2, 3, 4, 5)
# Create matrix
mat <- matrix(c(1,2,3,4,5,6,7,8,9), nrow = 3, byrow = FALSE)
# Create another list
inner_list <- list("a1", "a2", "a3")

# Combine into one list
my_list <- list(vec, mat, inner_list)
# Give names to elements
names(my_list) <- c("Vector", "Matrix", "List"
print("Complete List:")
print(my_list)
# Access 1st element
print("First element (Vector):")
print(my_list[[1]])
# Access 2nd element
print("Second element (Matrix):")
print(my_list[[2]])
```

Q2. Solve the following questions. Write a R program to plot a bar chart for the following data.
Expenses Dataset Sources Food Clothing Rent Bills Other Expense
2000 2500 5000 2000 1500
Expenses Dataset Sources Food Clothing Rent Bills Other

Sources	**Food**	**Clothing**	**Rent B**	**Bills**	**Other**
Expenses	**2000**	**2500**	**5000**	**2000**	**1500**

Apply appropriate styling and add labels to the graph.

Ans->

```
# Data
sources <- c("Food", "Clothing", "Rent", "Bills", "Other")
expenses <- c(2000, 2500, 5000, 2000, 1500)
# Create bar chart
barplot(expenses,
        names.arg = sources,
        col = c("red", "blue", "green", "orange", "purple"),
        main = "Monthly Expenses",
        xlab = "Expense Categories",
        ylab = "Amount (Rs)",
        border = "black")
# Add values on top of bars
text(x = seq_along(expenses),
     y = expenses,
     label = expenses,
     pos = 3)
```

Q9. Q1. Solve the following questions. A. Write a R program to call the (built-in) dataset air quality. Remove the variables 'Solar.R' and 'Wind' and display the data frame.

B.Create the following matrices in R and perform the following operations:

U = **23 52 71**

12 24 81

33 63 99

V= 1 4 7

2 5 8

3 6 9

A. Addition (U + V)

B. Subtraction (U – V)

Ans-> # Load built-in dataset

```
data("airquality")
# Display original data
print("Original Data:")
print(head(airquality))
# Remove 'Solar.R' and 'Wind'
new_data <- airquality[, !(names(airquality) %in% c("Solar.R", "Wind"))]
# Display updated data
print("Updated Data (without Solar.R and Wind):")
print(head(new_data))
# Create Matrix U
U <- matrix(c(23,12,33,
```

```
             52,24,63,
             71,81,99),
           nrow = 3, byrow = FALSE)
# Create Matrix V
V <- matrix(c(1,2,3,
             4,5,6,
             7,8,9),
           nrow = 3, byrow = FALSE)

# Display matrices
print("Matrix U:")
print(U)
print("Matrix V:")
print(V)
# A. Addition
add_result <- U + V
print("U + V:")
print(add_result)
# B. Subtraction
sub_result <- U - V
print("U - V:")
print(sub_result)
```

B.Solve the following questions. Write a R program to plot a bar chart for the following data. Student Dataset

Courses	**C++**	**Java**	**Python**	**DataScience**	**SQL**
% Obtained	**52**	**85**	**87**	**95**	**93**

Ans->

```
# Data
courses <- c("C++", "Java", "Python", "DataScience", "SQL")
marks <- c(52, 85, 87, 95, 93)
# Create bar chart
barplot(marks,
        names.arg = courses,
        col = c("red", "blue", "green", "purple", "orange"),
        main = "Student Performance",
        xlab = "Courses",
        ylab = "Percentage (%)",
        ylim = c(0, 100),
        border = "black")
# Add values on bars
text(x = seq_along(marks),
```

```
    y = marks,
    label = marks,
    pos = 3)
```

Q10. Solve the following questions. Write a R program to list containing a vector, a matrix and a list and give names to the elements in the list.
Print the list Vector : [1, 2, 3, 4, 5]
Matrix : 1 4 7
2 5 8
3 6 9
List : (a1, a2, a3)
Ans->

```
# Create vector
vec <- c(1, 2, 3, 4, 5)
# Create matrix
mat <- matrix(c(1,2,3,4,5,6,7,8,9), nrow = 3, byrow = FALSE)
# Create list
inner_list <- list("a1", "a2", "a3"
# Combine into one list
my_list <- list(vec, mat, inner_list)
# Give names to elements
names(my_list) <- c("Vector", "Matrix", "List")
# Print the list
print("Complete List:")
print(my_list)
```

Q2. Solve the following questions. Write a R program to plot a bar chart for the following data.

Courses	GO	Java	SQL	Hadoop	DataScience
% Obtained	58	75	89	85	63

Apply appropriate styling and add labels to the graph.
Ans->

```
# Data
courses <- c("GO", "Java", "SQL", "Hadoop", "DataScience")
marks <- c(58, 75, 89, 85, 63)

# Create bar chart
barplot(marks,
      names.arg = courses,
      col = c("skyblue", "orange", "green", "purple", "red"),
      main = "Student Performance",
      xlab = "Courses",
      ylab = "Percentage (%)",
```

```
        ylim = c(0, 100),
        border = "black")

# Add values on bars
text(x = seq_along(marks),
     y = marks,
     label = marks,
     pos = 3)
```

Q11. Q1. Solve the following questions. Create the following matrices in R and perform the following operations:

$$\boldsymbol{U} = \begin{matrix} \mathbf{11} & \mathbf{44} & \mathbf{77} \\ \mathbf{22} & \mathbf{55} & \mathbf{88} \\ \mathbf{33} & \mathbf{66} & \mathbf{99} \end{matrix}$$

$$\boldsymbol{V} = \begin{matrix} \mathbf{1} & \mathbf{4} & \mathbf{7} \\ \mathbf{2} & 5 & \mathbf{8} \\ \mathbf{3} & \mathbf{6} & \mathbf{9} \end{matrix}$$

A. Addition (U + V)
B. Subtraction (U – V)
C. Multiplication (UV)
Ans->

```
# Create Matrix U
U <- matrix(c(11,22,33,
              44,55,66,
              77,88,99),
            nrow = 3, byrow = FALSE)
# Create Matrix V
V <- matrix(c(1,2,3,
              4,5,6,
              7,8,9),
            nrow = 3, byrow = FALSE)
# Display matrices
print("Matrix U:")
print(U)
print("Matrix V:")
print(V)
# A. Addition
add_result <- U + V
print("U + V:")
print(add_result)
# B. Subtraction
sub_result <- U - V
print("U - V:")
```

```
print(sub_result)
# C. Multiplication
mul_result <- U %*% V
print("U %*% V:")
print(mul_result)
```

Q2. Solve the following questions. Write a R program to plot a bar chart for the following data. Expenses Dataset

Sources	**Food**	**Clothing**	**Rent**	**Bills**	**Other**
Expense	**2000**	**2500**	**5000**	**2000**	**1500**

Apply appropriate styling and add labels to the graph.

Ans->

```
# Data
sources <- c("Food", "Clothing", "Rent", "Bills", "Other")
expenses <- c(2000, 2500, 5000, 2000, 1500)
# Create bar chart
barplot(expenses,
      names.arg = sources,
      col = c("red", "blue", "green", "orange", "purple"),
      main = "Monthly Expenses",
      xlab = "Expense Categories",
      ylab = "Amount (Rs)",
      ylim = c(0, 6000),
      border = "black")
# Add values on top of bars
text(x = seq_along(expenses),
    y = expenses,
    label = expenses,
    pos = 3)
```

Q.12 Q1. Solve the following questions. Write an R program to extract first 10 English letter in lower case and last 10 letters in upper case and extract letters between 22nd to 24th letters in upper case. Print your name in upper case.

Ans->

```
# Extract first 10 English letters in lower case
first10_lower <- letters[1:10]
print("First 10 lowercase letters:")
print(first10_lower)
# Extract last 10 letters in upper case
last10_upper <- LETTERS[(length(LETTERS)-9):length(LETTERS)]
print("Last 10 uppercase letters:")
print(last10_upper)
# Extract letters between 22nd to 24th letters in upper case
```

```
letters_22_24 <- LETTERS[22:24]
print("22nd to 24th letters in uppercase:")
print(letters_22_24)
# Print your name in uppercase
name <- "YourName"  # Replace with your actual name
name_upper <- toupper(name)
print("Name in uppercase:")
print(name_upper)
```

Q2. Solve the following questions. A. Write a R program to find the mean of numbers from 84 to 95 B. Write a R program to find the variance of numbers from 45 to 105.
Ans->
mean of numbers from 84 to 95

```
# Create sequence
nums <- 84:95
# Calculate mean
mean_value <- mean(nums)
print(paste("Mean is:", mean_value))
```

variance of numbers from 45 to 105.

```
# Create sequence
nums <- 45:105
# Calculate variance
variance_value <- var(nums)
print(paste("Variance is:", variance_value))
```